A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

MARIO ASIS

ABSTRACT. This paper proposes that what appears of a thing, in the elementary case, is a difference that it bears, and that such a difference appears just where something is as it is in virtue of it. It offers these two conditions, differentiation and receptive dependence, as the minimal core of perception and of whatever knowledge rests on it, since nothing is perceived that does not appear. It states them between actual terms only, without alternatives, order, representation, or a knowing subject, and calls the resulting structure registration. What a subject adds lies outside the core.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper asks what must hold before that: what is required for anything to appear at all.

Its answer is a proposal in two parts. In the elementary case, what appears of a thing is a difference that it bears: nothing appears where everything is alike. And such a difference appears just where something is as it is in virtue of it. The paper calls this structure *registration*.

To appear is to show up. The word is not used in the sense of seeming, as opposed to being, and it implies no subject to whom something appears. Perception and knowledge, which involve a subject, are not treated here; nothing is perceived that does not appear, so on this account of appearing the conditions are necessary for them. The conditions are formal: they are stated without reference to space, time, matter, or causation, and examples from particular domains show how they are met there without fixing what they are.

The exposition uses ordinary logical resources—identity, distinction, relation, and inference—to state its commitments. This does not attribute those resources to what the commitments describe, and in particular not to a knowing subject. A distinction drawn in the exposition is not thereby a distinction drawn by its object.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from these in three respects. What is depended on is always a difference between actual terms. The dependence is stated without probabilities, causes, or alternatives. And its direction is distinguished from any order of occurrences.

Sections 2 and 3 concern the difference in what is under consideration. Sections 4 and 5 concern the dependence and the receiving side. Section 6 defines registration, Section 7 gives two cases, and Section 8 concerns order.

2. Differentiation

Let F designate what is under consideration. Calling it a *field* fixes the scope of the inquiry; it implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. Nor does the notation establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification.

The account takes difference as primitive. The notion is not defined, since any definition would already use it: to define something is to set it apart from something else. It can be pointed to. Take anything that is not alike throughout. Something in it is one way, and something in it is another way: there are two, and in some respect they are not alike. The account considers differences in this form, between two actual terms in a given respect, and it calls a difference of this form a *contrast*. The two that differ are its *terms*, that in which they differ is its *respect*, and what has the two terms *bears* the contrast. Where more than two terms differ, the account considers them two at a time. Two contrasts are the same when their terms and their respect are the same, so the same two terms may bear two contrasts, in two respects. A contrast belongs to what is under

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither.

consideration; the articulation by which an account speaks of it is introduced in Section 3.

(D) DIFFERENTIATION

What appears of F, in the elementary case, is a contrast that F bears.

The account takes appearing to be contrastive: nothing appears where everything is alike, and what appears stands out from something. If F is taken whole, what appears is its difference from something beside it, and what is under consideration is then the two together. If F is everything, there is nothing beside it, and what appears is the differences within it. "In the elementary case" leaves room for what Section 6 says of anything else that appears: that it appears through contrasts, not that it is one.

Both terms of a contrast are actual. The account uses no difference between something actual and an alternative, that is, a way that nothing actual is. This is a commitment of the account. To use alternatives it would have to say where they are held, in a stock of possibilities, in a repertoire belonging to whatever receives, or in the description that specifies them, or else take on a modal commitment about the thing itself. It takes on none of these. The core therefore does not provide for three things to appear: which way something is as against a way that nothing is; anything of a term that differs from nothing actual; and that two terms are alike. "Actual" is meant without tense.

That F bears a contrast says that there is a difference, not that F could be differentiated. It says nothing of which contrasts F bears, of whether one differentiation of F is privileged, or of whether what differs is objects, qualities, regions, or values. It implies nothing about change or order. And it involves no knowing subject: that F bears a contrast does not imply that anything differentiates it.

The contrasts concerned are those of F itself. The account concerns a differentiation attributable to its target, rather than one supplied entirely by an unrelated description. A difference that holds, even in part, because the account has specified it belongs to the description, and it is left out. A respect that the account chooses is not thereby one that it supplies: what matters is whether the terms differ in it apart from the choosing.

3. Articulation

An account speaks of a particular contrast of F by means of an *articulation*, written

$$\langle a, \delta, b \rangle_F,$$

which names two terms, a and b, and a respect, δ. The articulation is *specified* by the account, not by F. It is *realized* when a and b are actual terms of F and differ in δ. Specification selects a contrast; it does not create one, and it can fail: a name may pick out nothing, or the terms may not differ in the respect named. The order in which the terms are written, their labels, and their grouping belong to the articulation, not to F. Specification fixes what an inquiry concerns, as F fixes its scope; it does not supply what the inquiry finds.

The terms of a realized articulation are both actual. They are not alternatives of which one is the case and the other not; the contrast is the difference between two terms that are both actual. The schema does not assume that a and b exist before the contrast as independent objects.[2]

Two misreadings of the schema should be excluded. First, δ is not a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ is the respect in which the two differ. Second, b need not comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the field": the latter adds the requirement that every case within the scope fall on one side or the other (Figure 1).

A realized articulation remains an articulation of F. Nothing so far implies that its contrast appears anywhere.

4. Receptive Dependence

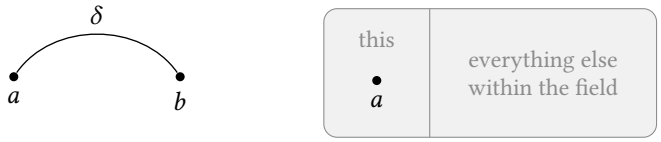
That F bears a contrast, and that something else is in some condition, are two facts, and nothing said so far connects them. They could be unrelated. A further commitment is therefore needed.

> (C) RECEPTIVE DEPENDENCE
>
> *A contrast of F appears only where some condition of something holds, at least partly, in virtue of it.*

A *condition* is something's being a certain way. (C) takes (D) for granted: what appears is a contrast. What it adds is the dependence.

"In virtue of" is not defined. It is the notion in use whenever one thing is given as the reason why another is so. Two things may both be the case without

2. That a difference need not be grounded in terms identified before it is argued by Deleuze in his critique of the subordination of difference to identity (Deleuze 1994, ch. 1); the present account leaves the question open. Deleuze also rejects the possible as a mere image of the real, and the present account likewise uses no stock of possibilities.



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark terms, both actual, and δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

either having anything to do with the other; or one may be the case because the other is. (C) says that appearing is of the second kind. It is receptive: where a contrast of F appears, something is as it is because of it. The account also calls this *dependence*. In what follows K is a contrast of F and q a condition that holds in virtue of it.

The notion is used under three conditions. It relates what holds: K and q both hold. Nothing holds in virtue of itself. And it has a direction: that q holds in virtue of K does not bring with it that K holds in virtue of q.[3]

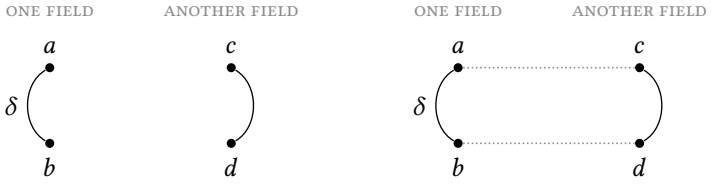
The dependence is on the difference itself, not merely on how one of its terms is. A condition that held in virtue of how a is alone would not depend on the contrast of a and b. This does not set the contrast against how the two terms are together: the contrast is the respect in which what they jointly are matters to the condition.

Dependence is neither resemblance nor pairing. Something elsewhere may be like the contrast, and its parts may be paired with the terms of the contrast, without holding in virtue of it (Figure 2).[4] Both relations can be read in either direction, and neither ties one thing to this contrast rather than to any that is like it. Nor does agreement show dependence: where two things agree because both depend on a third, their agreement does not establish that either depends on the other. (C) requires none of these, and in particular no pairing.

Nor is dependence the same as what would have held otherwise. q may hold in virtue of K although, had K not held, something else would have brought q about; and it can be that neither K nor q holds without the other although both depend on a third. What would have held otherwise is evidence of dependence,

3. The first and third features are familiar from the literature on metaphysical ground (Fine 2012). The account does not identify the dependence with ground.

4. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot be what representation consists in.



(a) resemblance (b) pairing

FIGURE 2. Two relations that are not dependence. (a) Resemblance: two contrasts are alike. (b) Pairing: a is paired with c, and b with d. The pairing is drawn without direction, since it can be read either way.

not what it consists in.[5]

The dependence may be partial: q may hold in virtue of K and of other things as well, among them how its bearer is.[6] Bateson's definition of information as "a difference which makes a difference" is close to this (Bateson 1972); MacKay had earlier described information as a distinction that makes a difference (MacKay 1969). The difference is K, and its making a difference is q's holding in virtue of it. The dependence should not be replaced by an appeal to "systematic" or "relevant" coupling, which brings in a standard of relevance or a regularity across cases.

What the dependence consists in is a matter for each domain: in physical domains it may be causal, in others logical or constitutive. The core requires that there be such a dependence, not that it be of one kind. It is admitted, not derived from what it relates: a relation cannot be obtained by describing what it relates. Whether it is transitive is left open.

A condition may depend on several contrasts. It then stands in the same relation to each, and the core does not single out one of them.

The direction of the dependence gives two roles. In an instance of it, what bears the contrast is the *target*, and what bears the dependent condition is the *receiver*. The roles are places in the relation, not natures of what occupies them: the same thing may be target in one instance and receiver in another.

5. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts hold necessarily, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997).
6. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).

5. The Receiving Side

Let R designate what bears the dependent condition.[7] R need not be a person, a conscious subject, or an independently bounded system, and it may be an aspect of a larger organization that also includes F, since a distinction of roles does not establish two substances.

The receiving side needs one condition, not two. The condition is actual, and it need not be like the contrast of F in any way; nothing in it is assigned to one term of the contrast rather than the other. No further contrast borne by R is required for it to register a contrast of F. In particular, R need not be in a second condition from which the first differs.

The account attributes no act of distinguishing to R. That its condition holds in virtue of a contrast of F is a fact about the contrast and the condition, not something that R has to note. It is the account that picks them out.

The condition of R can itself appear only as (D) allows: where it differs from something actual, and that contrast is depended on in turn.

Neither side comes with the other. A contrast of F may hold while nothing depends on it, and R may be in conditions that depend on nothing in F.

This fixes how much individuation the account admits. There must be enough to distinguish the contrast from the condition that depends on it. There need not be enough to identify enduring objects or successive states of a continuing subject. To assume that would build into the foundation a persisting, bounded bearer—which is what an account of a knowing subject must eventually explain.

6. Registration

DEFINITION OF REGISTRATION

A contrast K is registered where a condition q of something holds, at least partly, in virtue of K. What bears q registers K.

Registration is always of a particular contrast: to say that R registers F without qualification is to leave the claim incomplete. It is an actual relation between a contrast and a condition that both hold, not a disposition of R to respond (Figure 3).

The account proposes registration as the elementary form of appearing. Every registration is an appearing. And whatever else appears is claimed to

7. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.

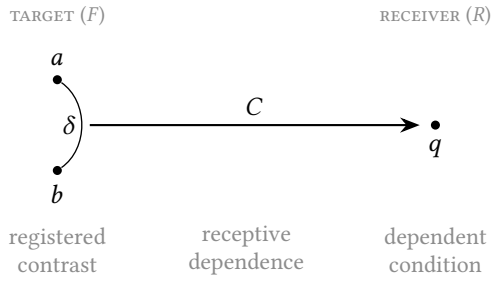


FIGURE 3. Registration. A condition q of R holds at least partly in virtue of the contrast of a and b in δ. The arrow runs from the contrast, not from either term, and denotes receptive dependence only: it represents neither a transfer nor an order of occurrences. The columns are roles, not regions.

appear through registrations that are part of it, not merely alongside it. The second claim is not shown here: it states what an account of the three things set aside in Section 2 would have to meet.

The proposal is broad, and deliberately so. A contrast appears wherever it contributes to why some condition holds. The condition need not display the contrast, distinguish it from others, or keep anything of it.

Registration is supplied by neither side alone. There is the contrast, the condition, and the dependence of the second on the first. Keeping them distinct prevents registration from being attributed to the target merely because the target is differentiated, or to the receiver merely because it is in some condition.

The definition attributes to the receiver no understanding, identification, or judgement. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place anything "in the logical space of reasons".[8] It requires no transfer from F to R, no resemblance, and no representational space or repertoire in R.

What is registered is always a difference between actual terms: that a differs from b. Accounts of information and knowledge that begin from alternatives presuppose a set among which the actual case is singled out;[9] the core has no such set.

8. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, contains a registration in the present sense (Dretske 1969).

9. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives (Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

7. Two Cases

In the first case the dependence is causal. Let a and b be two bodies of unequal mass, resting on the pans of a balance, and let the beam be inclined. K is the contrast of the bodies in mass, and q is the beam's being inclined. q holds in virtue of K, in part: the beam is inclined because the two sides are unequally loaded, and not because of the mass of either body considered alone. How far it is inclined depends on more: on how much the masses differ, on how the balance is built, and on where the bodies rest. The balance registers the difference in mass. It has no repertoire of the masses that the bodies might have had.

In the second the dependence is constitutive. Let a and b be two parts of a whole W that differ in a respect δ. K is their contrast, and q is the whole's being non-uniform in δ. q holds in virtue of K: W is non-uniform because two of its parts differ. Nothing causes anything here, and nothing passes from one thing to another. The whole registers a contrast within itself, through a condition of its own.

Nobody would speak of perception or knowledge in either case: those involve a subject, which the core does not contain.

8. Registration and Order

The core assumes no order. The dependence has a direction, but the direction of a relation is not an ordering of occurrences: that q holds in virtue of K does not place either before the other.

The core therefore cannot say that a receiver *acquires* what it registers, *retains* it, or *compares* it with something else.[10] Retention requires that one receiver be identified across distinct places of some ordering. Acquisition requires that a place at which a contrast is not registered stand in an asymmetric relation to one at which it is. Neither follows from (C).

What they require is at least an asymmetric relation among places, with no requirement that any two places be comparable, and with no metric, no linearity, and no universal index. Such a relation may still run in a circle, so it is less than an order in the usual sense, and what more is needed is left open. Temporal succession is one relation of this kind. Whether a relation that is not

10. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account speaks only of order, of which temporal order is one kind.

temporal, such as the order of steps in a proof, could serve for acquisition or retention is also left open.

If receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The direction of dependence and the asymmetry of an ordering play different roles and should be kept notationally distinct.

9. Conclusion

Table 1 summarizes the elements of the account.

Table 1. The elements of the account, with what each permits and what each leaves unestablished.

element	what it permits	what it does not establish
(D) Differentiation	A difference attributable to what is under consideration, which is what appears of it.	Which contrasts F bears; that they appear anywhere.
Actual terms only	A core without possibilities or a repertoire.	Which way a term is as against an alternative; anything of a lone term; that two terms are alike.
Specified articulation	Terms and the respect in which they differ.	Its own realization, an exhaustive partition, or objecthood.
(C) Receptive dependence	Registration of a contrast; the roles of target and receiver.	Understanding, representation, order, acquisition, or retention.
The receiver R	A condition that depends on a contrast.	A knowing subject; a second condition; any resemblance to the contrast.

The proposal is that what appears of a thing, in the elementary case, is a contrast that it bears, and that a contrast appears just where something is as it is in virtue of it, with target and receiver distinguished as roles in that dependence.

Five questions remain open. What the dependence consists in within a given domain, and whether it can be explained by more elementary relations; to write it as an arrow would not answer this. What must be added to a registration for something to appear to a subject, that is, to be perceived. How which way a term is, and that two terms are alike, can appear without alternatives. What acquisition and retention require beyond an asymmetric relation. And whether what is said of things that cannot act on whoever speaks of them, things picked

out by description, future events, and mathematical objects, rests on registration at all.

The account ends at registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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